Program Check Report

Static review of saved program source against published UR guidance. No program was executed and no robot was commanded. Dynamic values, execution order and physical correctness are not inferred.

PROGRAM SUMMARY	
Program	App_DoughFactory
Software saved in URP	5.24.0
Guidance profile	CB-Series / e-Series deployment guidance
Script lines / motion calls	1,920 lines; 27 movej, 35 movel, 0 movep
Other motion commands	9 optimovej
Payload / TCP source calls	8 payload; 2 TCP
Generated	2026-10-29 19:08 UTC
Overall result	Review required — source checks do not certify deployment

SCOPE & INTERPRETATION

UR deployment acceleration values are suggested starting points for CB-Series/e-Series, not specifications or safety limits. Confirm the selected profile, physical payload/TCP, controller version and application before acting.

Only supplied source is reviewed. Variables, called helper functions, branches, installation data, waypoint transforms, URCap behavior and actual motion are not simulated. Source call counts include definitions and do not indicate execution frequency.

Independent source review by UR Motion Studio. Not a Universal Robots certification or safety validation. Runtime behavior, physical setup and application suitability require verification on the applicable robot.

Review checklist

CHECKLIST	
Initial payload / CoG / inertia	Source observation
Dynamic payload changes	Manual review
Initial TCP	Source observation
Dynamic TCP changes	Manual review
Acceleration recommendations	Review recommended
Blend radius overlap	Manual review
Blended motion continuation	Manual review

READ THE RESULTS

Within guideline: a resolvable classic motion acceleration is within the cited starting value. Review recommended: a concrete source value deserves attention. Source observation: a configuration call exists. Manual review: the evidence cannot establish the runtime or physical result.

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Initial payload / CoG / inertia

Source observation · Line 11

First source occurrence; execution order and physical correctness are not proven. Mass: 3.6 kg; cog: [0.001, 0.0, 0.132] m from tool mount; inertia: [0.013017, 0.013017, 0.013017, 0.0, 0.0, 0.0] kg·m². Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_target_payload(3.600000, [0.001000, 0.000000, 0.132000], [0.013017, 0.013017, 0.013017, 0.000000, 0.000000, 0.000000])

UR reference [5]

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Dynamic payload changes

Source observation · Line 514

Additional source occurrence; execution order and physical correctness are not proven. Mass: dynamic or not supplied; cog: dynamic or not supplied; inertia: [0, 0, 0, 0, 0, 0] kg·m². Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_target_payload(PAYLOADS.FULL.M, PAYLOADS.FULL.COG, transition_time=0.25)

UR reference [5]

Source observation · Line 799

Additional source occurrence; execution order and physical correctness are not proven. Mass: dynamic or not supplied; cog: dynamic or not supplied; inertia: [0, 0, 0, 0, 0, 0] kg·m². Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_target_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG, transition_time=0.25)

UR reference [5]

Source observation · Line 1535

Additional source occurrence; execution order and physical correctness are not proven. Mass: dynamic or not supplied; cog: dynamic or not supplied. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG)

UR reference [1]

Source observation · Line 1725

Additional source occurrence; execution order and physical correctness are not proven. Mass: dynamic or not supplied; cog: dynamic or not supplied. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.FULL.M, PAYLOADS.FULL.COG)

UR reference [1]

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Detailed findings (continued)

Source observation · Line 1731

Additional source occurrence; execution order and physical correctness are not proven. Mass: dynamic or not supplied; cog: dynamic or not supplied. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG)

UR reference [1]

Source observation · Line 1811

Additional source occurrence; execution order and physical correctness are not proven. Mass: dynamic or not supplied; cog: dynamic or not supplied. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.FULL.M, PAYLOADS.FULL.COG)

UR reference [1]

Source observation · Line 1817

Additional source occurrence; execution order and physical correctness are not proven. Mass: dynamic or not supplied; cog: dynamic or not supplied. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

set_payload(PAYLOADS.EMPTY.M, PAYLOADS.EMPTY.COG)

UR reference [1]

Manual review

8 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. Confirm tool plus workpiece mass, centre of gravity and inertia against the real tool; update payload when the workpiece changes.

UR reference [1]

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Initial TCP

Source observation · Line 9

First source occurrence; execution order and physical correctness are not proven. pose: [0.0, 0.0, 0.247, -0.0, 3.1414, -0.0] (XYZ in m; rotation vector in rad). Confirm the configured TCP against the actual tool and each tool change.

```
set_tcp(p[0.0,0.0,0.247,-0.0,3.1414,-0.0])
```

UR reference [1]

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Dynamic TCP changes

Source observation · Line 1536

Additional source occurrence; execution order and physical correctness are not proven. pose: dynamic or not supplied. Confirm the configured TCP against the actual tool and each tool change.

set_tcp(TCPS.UNDER_FLANGE)

UR reference [1]

Manual review

2 configuration call(s) in source, including definitions and branches. Call counts do not prove updates at every pickup, release or tool change. Confirm the configured TCP against the actual tool and each tool change.

UR reference [1]

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Acceleration recommendations

Classic MoveJ/MoveL/MoveP: compare configured acceleration with UR starting recommendations [1], only when t=0 and the robot family is confirmed. Dynamic expressions and other commands remain unassessed. Duration overrides acceleration [2].

Review recommended · Line 448

Configured a=3 m/s²; UR suggested starting value ≤2.5 m/s². This is not a safety limit or measured acceleration.

movel(retractTo, a=3.0, v=0.1)

UR reference [1]

SOURCE PARAMETER EVIDENCE	
Line 138	Manual review · movej · a=ar3*a_scale rad/s²; t=0
Line 167	Within guideline · movel · a=1 m/s²; t=0
Line 199	Within guideline · movel · a=0.05 m/s²; t=0
Line 327	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 329	Manual review · movej · a=ar1*a_scale rad/s²; t=0
Line 360	Manual review · movel · a=al2*a_scale m/s²; t=0
Line 390	Within guideline · movel · a=0.2 m/s²; t=0
Line 414	Manual review · movel · a=al3*a_scale m/s²; t=0
Line 427	Within guideline · movel · a=1.2 m/s²; t=0
Line 482	Within guideline · movel · a=0.1 m/s²; t=0
Line 577	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 591	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 599	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 616	Manual review · movel · a=al2*a_scale m/s²; t=0
Line 618	Manual review · movej · a=ar3*a_scale rad/s²; t=0

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SOURCE PARAMETER EVIDENCE (continued)	
Line 623	Manual review · movel · a=al2*a_scale m/s²; t=0
Line 637	Manual review · movel · a=al2*a_scale m/s²; t=0
Line 644	Manual review · movel · a=al2*a_scale m/s²; t=0
Line 657	Manual review · movej · a=ar3*a_scale rad/s²; t=0
Line 664	Manual review · movej · a=ar3*a_scale rad/s²; t=0
Line 672	Manual review · movej · a=ar3*a_scale rad/s²; t=0
Line 679	Manual review · movej · a=ar3*a_scale rad/s²; t=0
Line 707	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 711	Manual review · movej · a=ar2*a_scale rad/s²; t=0
Line 737	Manual review · movel · a=vl2*a_scale m/s²; t=0
Line 741	Manual review · movel · a=vl2*a_scale m/s²; t=0
Line 744	Manual review · movel · a=al2*a_scale m/s²; t=0
Line 761	Within guideline · movel · a=0.3 m/s²; t=0
Line 797	Manual review · movel · a=ar2*a_scale m/s²; t=0
Line 807	Manual review · movel · a=ar2*a_scale m/s²; t=0
Line 826	Manual review · movel · a=ar2*a_scale m/s²; t=0
Line 828	Manual review · movel · a=ar2*a_scale m/s²; t=0
Line 831	Manual review · movej · a=vr2*a_scale rad/s²; t=0
Line 843	Manual review · movel · a=ar1*a_scale m/s²; t=0
Line 858	Manual review · movel · a=al3*a_scale m/s²; t=0
Line 865	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1471	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1473	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1476	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0

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SOURCE PARAMETER EVIDENCE (continued)	
Line 1478	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1481	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1483	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1485	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1487	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1490	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1492	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1494	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1496	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1498	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1500	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1502	Within guideline · movej · a=1.3962634015954636 rad/s²; t=0
Line 1747	Manual review · movel · a=a_scale*al2 m/s²; t=0
Line 1758	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 1765	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 1782	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 1787	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 1791	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 1793	Manual review · movel · a=al1*a_scale m/s²; t=0
Line 1801	Manual review · movej · a=ar3 rad/s²; t=0
Line 1803	Manual review · movej · a=ar3 rad/s²; t=0
Line 1835	Manual review · movel · a=ar1*a_scale m/s²; t=0
Line 156	Manual review · optimovej · a=0.8 command-specific units; t=n/a
Line 158	Manual review · optimovej · a=0.8 command-specific units; t=n/a

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SOURCE PARAMETER EVIDENCE (continued)	
Line 161	Manual review · optimovej · a=.6 command-specific units; t=n/a
Line 294	Manual review · optimovej · a=1 command-specific units; t=n/a
Line 296	Manual review · optimovej · a=1 command-specific units; t=n/a
Line 303	Manual review · optimovej · a=1 command-specific units; t=n/a
Line 307	Manual review · optimovej · a=.5 command-specific units; t=n/a
Line 639	Manual review · optimovej · a=1 command-specific units; t=n/a
Line 1806	Manual review · optimovej · a=0.5 command-specific units; t=n/a

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Blend radius overlap

Configured blend radii are source observations [4]. No overlap or skipped waypoint is proven: target poses, transforms, control flow and controller motion version must be verified. Version 2 can adapt blend radii [3].

Manual review
line 13: motion_version_set(2) Version 2 can reduce blend radii during planning. A static geometric overlap must not be presented as a proven skipped waypoint. UR reference [3]

SOURCE PARAMETER EVIDENCE	
Line 138	movej · r=0.02 m · Manual review
Line 329	movej · r=0.01 m · Manual review
Line 599	movel · r=0.003 m · Manual review
Line 616	movel · r=0.08 m · Manual review
Line 618	movej · r=0.1 m · Manual review
Line 623	movel · r=0.05 m · Manual review
Line 637	movel · r=0.08 m · Manual review
Line 644	movel · r=0.05 m · Manual review
Line 657	movej · r=0.05 m · Manual review
Line 664	movej · r=0.05 m · Manual review
Line 672	movej · r=0.05 m · Manual review
Line 679	movej · r=0.05 m · Manual review
Line 737	movel · r=0.05 m · Manual review
Line 741	movel · r=0.05 m · Manual review
Line 807	movel · r=0.0025 m · Manual review
Line 826	movel · r=0.0025 m · Manual review

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SOURCE PARAMETER EVIDENCE (continued)	
Line 828	movel · r=0.0025 m · Manual review
Line 831	movej · r=0.02 m · Manual review
Line 843	movel · r=0.01 m · Manual review
Line 858	movel · r=0.05 m · Manual review
Line 1492	movej · r=0.05 m · Manual review
Line 1758	movel · r=0.05 m · Manual review
Line 1765	movel · r=0.05 m · Manual review
Line 1782	movel · r=0.02 m · Manual review
Line 1787	movel · r=0.02 m · Manual review
Line 1791	movel · r=0.02 m · Manual review
Line 1793	movel · r=0.01 m · Manual review
Line 1801	movej · r=0.1 m · Manual review
Line 1803	movej · r=0.1 m · Manual review

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Blended motion continuation

Manual review

29 classic motion calls have nonzero or unresolved radii. Verify continuation through branches, waits, stops and helper functions on the applicable controller. Static source inspection does not establish the executed path.

UR reference [4]

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Sources & next steps

Confirm matching URP/script versions. Review flagged lines in PolyScope, verify physical payload and TCP, and validate motion using the relevant robot/software manuals. Use flight measurements and incident logs to investigate observed behavior separately.

INPUT SOURCE

/home/apex/workspace/motionSim/assets/flightreports/robot_configuration20261014_19_16_56.zip :: files/programs/App_DoughFactory/App_DoughFactory.urp

SHA-256: cfaf558b332708eb69e1392c6a8f28c903802e4e70f393558cc058e9c34690a8

INPUT SOURCE

/home/apex/workspace/motionSim/assets/flightreports/robot_configuration20261014_19_16_56.zip :: files/programs/App_DoughFactory/App_DoughFactory.script

SHA-256: 6c79bdc83f15cc91e91158f1091f4119492ecc3e4d982f21edfb7503002ae790

UR REFERENCE [1]

UR Important Deployment Points (CB-Series/e-Series)

https://www.universal-robots.com/articles/ur/robot-care-maintenance/important-deployment-points/

UR REFERENCE [2]

UR PolyScope 5.24 Script Manual: MoveJ

https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/movej_qa14v105t0r.htm

UR REFERENCE [3]

UR Motion Version (PolyScope 5.22+)

https://www.universal-robots.com/manuals/EN/HTML/SW5_22/Content/prod-scriptmanual/all_scripts/motion_version_desc.htm

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Detailed findings (continued)

UR REFERENCE [4]

UR Blend Parameters

https://www.universal-robots.com/manuals/EN/HTML/SW5_19/Content/prod-usr-man/software/PolyScope/content/BasicProgNodes/commandtab_way_blend_para.htm

UR REFERENCE [5]

UR Payload configuration (PolyScope 5.24)

https://www.universal-robots.com/manuals/EN/HTML/SW5_24/Content/prod-scriptmanual/all_scripts/set_target_payload.htm

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